

SC 663: Optimization and Inference in Sparse Network Models

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Instructions for Students

1. **Using LaTeX:** LaTeX is a typesetting system used for technical and scientific documents. This template already includes the necessary packages for mathematical symbols, equations, and Python code. You can start writing your answers directly in this file.
2. **Using Overleaf:** You will turn in Overleaf is an online LaTeX editor. If you are new to Overleaf, create an account at <https://www.overleaf.com/> and start a new project by copying and pasting this template into the editor. A lot of tutorials are available on the internet—so, this will be easy.
3. **Inserting Python/MATLAB/Julia Code:** You can copy and paste code directly into this document using the following format:

```
1 # Example of a well-commented Python code
2 import numpy as np
3
4 def compute_square_root(x):
5     """
6     Function to compute the square root of a number
7     :param x: A non-negative number
8     :return: The square root of x
9     """
10    if x < 0:
11        raise ValueError("Cannot compute the square root of a negative number.")
12    return np.sqrt(x)
13
14 # Usage example
15 result = compute_square_root(9)
16 print("The square root of 9 is:", result)
```

Make sure to comment your code properly, explaining the logic and steps clearly. **If the code is long, just show only important snippets. Make sure to display your results.**

4. **Responsible Use of Gen-AI tools:** You may use large language models like ChatGPT or Claude for help in stylizing your code or assistance with mathematical concepts. Nonetheless you must acknowledge the use of Gen-AI tools. For example, I worked with ChatGPT for creating this template. That does not mean you give instructions for ChatGPT for solving your homework! Rather if you are stuck ask for hints. This is a math class so the more you think, the more intellectually you grow, and that growth is more satisfactory than 10% rate on your stocks! In your submission, please mention how and where you used it. Example: "Used ChatGPT to help improve the readability of Python code in Problem 2."
5. **Textbooks and References:** The first is the main textbook for this course.
 - Sparsity Methods for Systems and Control (SMSC): [Edition 1](#) and [Edition 2](#)
 - Sparse Estimation with Math and Python (SEMP): [link](#) - IITB Wifi or VPN needed.
 - Statistical Learning with Sparsity (SLS): [link](#)

Run the main.tex file

Create a Overleaf account. Create a project and execute main.tex file. If you want to run main.tex on your local computer, please install latex related packages.

HW 1: Sparse modeling and convex formulation

Problem 1. For $x = (x_1, \dots, x_n) \in \mathbb{R}^n$ and $1 \leq p < \infty$, define $\|x\|_p^p = \sum_{i=1}^n |x_i|^p$ (expressed slightly different than the book) and $\|x\|_\infty = \max_{1 \leq i \leq n} |x_i|$. Let $\|x\|_0$ denote the number of nonzero entries of x .

1. Prove that $\|x\|_\infty = \lim_{p \rightarrow \infty} \|x\|_p$.
2. Let $0 < p < 1$ in $\|x\|_p$. Show $\lim_{p \rightarrow 0^+} \|x\|_p^p = \|x\|_0$ (We have seen this result pictorially in class.)
3. Prove that $\|x + y\|_0 \leq \|x\|_0 + \|y\|_0$. When does equality hold?

Problem 2. Exercise 2.13 on page 26 in the [Edition 1](#) of the SMSC book.

Problem 3. Let $x^* \in \mathbb{R}^n$ be a sparsest solution of $\Phi x = y$, where $\Phi \in \mathbb{R}^{m \times n}$ with $m < n$. Suppose that $\text{supp}(x^*) = \{i_1, \dots, i_s\}$. Writing $\Phi = [\phi_1, \dots, \phi_n]$, show that the columns $\phi_{i_1}, \dots, \phi_{i_s}$ are linearly independent.¹

Problem 4 (Polynomial curve fitting in Ch 3 of the SMSC book). Define the $n - 1$ -th order polynomial:

$$y = f(t) = a_{n-1}t^{n-1} + \dots + a_1t + a_0.$$

Define $a = [a_0, a_1, \dots, a_{n-1}]^\top$ and $y = [y_1, \dots, y_m]^\top$. Consider the curve-fitting equations as $y = \Phi a$, where $\Phi \in \mathbb{R}^{m \times n}$ is the corresponding Vandermonde matrix.

1. Let $\Psi \in \mathbb{R}^{m \times n}$. Derive the solution of the weighted regularized least-squares problem

$$\min_x \frac{1}{2} \|\Phi x - y\|_2^2 + \frac{\lambda}{2} \|\Psi x\|_2^2, \quad \lambda > 0.$$

2. Describe the behavior of the regularized solution as $\lambda \rightarrow 0$ and as $\lambda \rightarrow \infty$.
3. Show that, for any $\Phi \in \mathbb{R}^{m \times n}$ and $\lambda > 0$, the matrices $\lambda I + \Phi \Phi^\top$ and $\lambda I + \Phi^\top \Phi$ are invertible. Prove that $\Phi^\top (\lambda I + \Phi \Phi^\top)^{-1} = (\lambda I + \Phi^\top \Phi)^{-1} \Phi^\top$.

Problem 5 (Computational exercise: polynomial fitting). Consider the following temperature measurements over a 24-hour period:

t	1	2	3	4	5	6	7	8	9	10	11	12
y	75	77	76	73	69	68	63	59	57	55	54	52
t	13	14	15	16	17	18	19	20	21	22	23	24
y	50	50	49	49	49	50	54	56	59	63	67	72

To avoid numerical difficulties associated with powers of t , first normalize the sampling instants as

$$s_i = \frac{t_i - 12.5}{11.5}, \quad i = 1, \dots, 24.$$

Consider a polynomial of order 23; that is $n - 1 = 23$. Using CVX (see [Edition 1](#) of SMSC for Matlab and [Edition 2](#) for Python), perform the following experiments.

1. Compute the interpolating polynomial, the ridge polynomial, and the sparse polynomial. For the latter two, use several values of $\lambda > 0$.
2. For representative values of λ , plot the temperature measurements together with the three fitted curves. Also plot the corresponding coefficient vectors x in another plot to visualize sparsity.
3. For a fitted coefficient vector x , define the relative fitting error as

$$E(x) = \frac{\|\Phi x - y\|_2}{\|y\|_2}.$$

Compare ridge and ℓ_1 regularization in terms of fitting error and sparsity of x as λ varies. Briefly explain your observations.

¹**Hint for Problem 3:** Assume the columns are linearly dependent and construct a nonzero vector d supported on $\text{supp}(x^*)$ such that $\Phi d = 0$. Then consider $x^* + \alpha d$ for a suitable choice of α .

Problem 6 (On Convexity). Exercises 4.14, 4.22, and 4.23 in [Edition 1](#).

Problem 7 (Proximal Operators). Exercises 4.19, 4.21, and 4.24 in [Edition 1](#).